Supplement 3: Additional Figures

Two survey galleries referred to from the main paper. Each shows a family of runs at a glance rather than carrying an argument on its own: the first sweeps the rotation offsets that Part I classifies, the second shows the spacetime fields behind the diversity mechanisms of Part II. They are collected here so the main narrative is not interrupted by four full pages of plates.

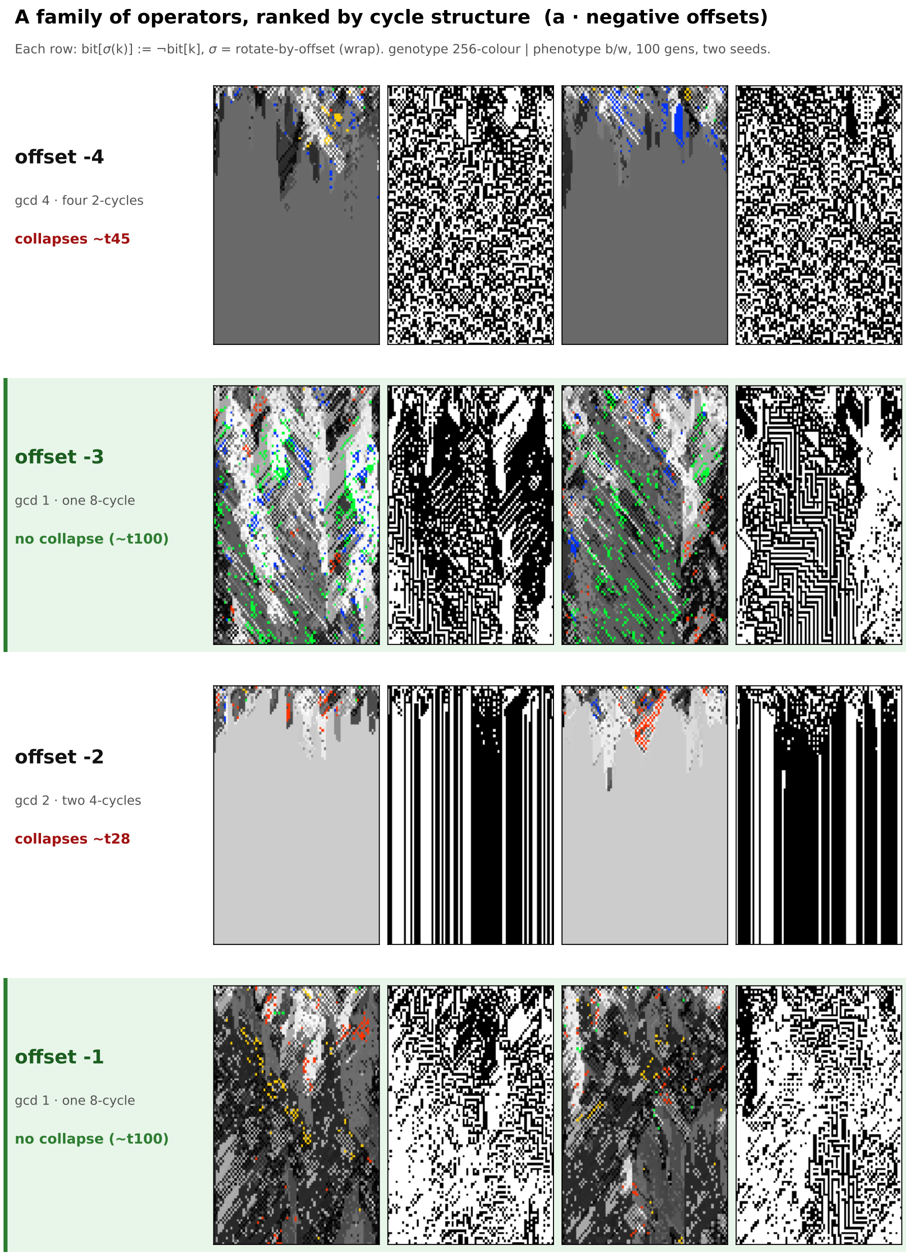

Figure S3.1

(a) Negative offsets. *The rotation subfamily, ordered by offset: each row is one wrap rotation* $\mathrm{bit}[\sigma(k)] := \neg\,\mathrm{bit}[k]$ *(two seeds; genotype in 256 colours, phenotype black and white, first hundred generations). Of these all-even rotations only the single 8-cycles,* $\gcd(\mathit{offset}, 8) = 1$*—offset* ± 1 *and* ± 3*, highlighted—sustain a high-diversity phase; offset* ± 2 *(*$\gcd 2$*) and* ± 4 *(*$\gcd 4$*) collapse. See Worked Worked example 4; continued in (b).*

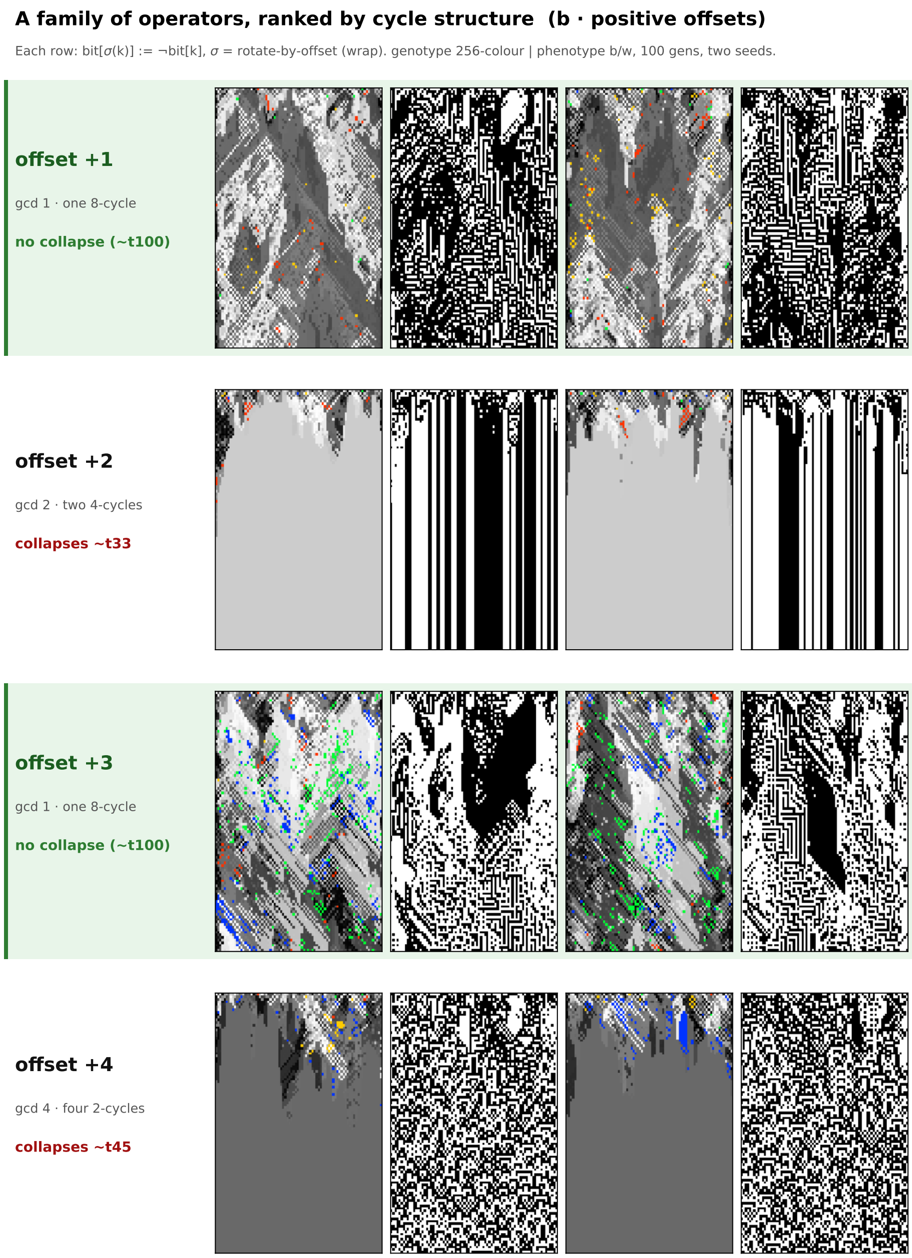

Figure S3.1

(b) Positive offsets. *Continued from (a). The classification is symmetric under $d \to -d$ (verdict follows $\gcd(d, 8)$), so offset $+1$ sustains as offset -1 does, and offset $+2$ collapses as offset -2 does—inverse rotations sharing verdict but not their transients (Worked Worked example 4). Possessing a fixed point and staying near it are different properties: the two pages visually summarise the distinction between fixed-point existence and spatial persistence.*

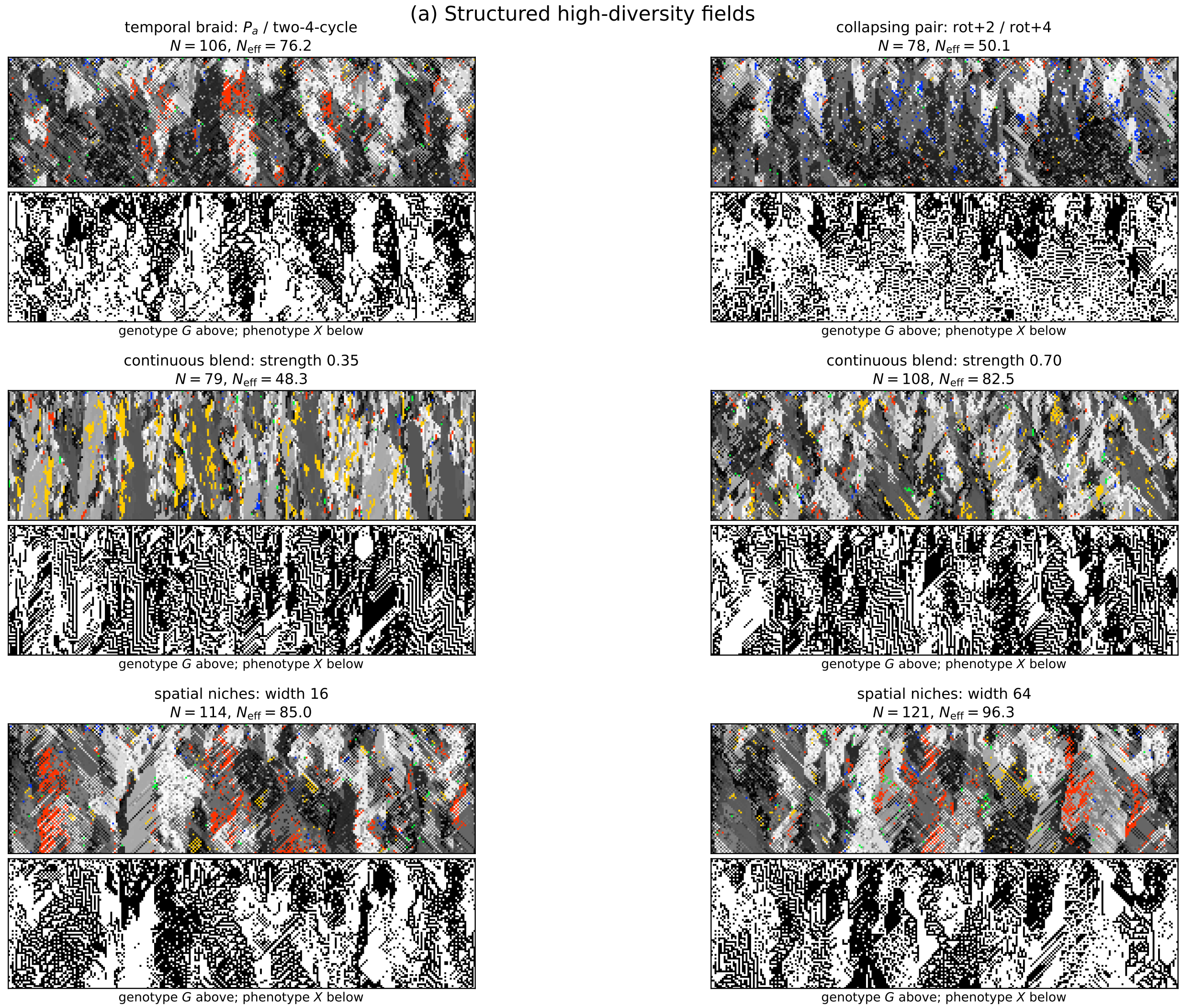

Figure S3.2

(a) Structured high-diversity fields. *Six examples from four mechanism families: temporal alternation; alternation of two operators that* collapse *when run alone; continuous blending at two strengths; and spatial niches at two widths. Each tile stacks genotype over phenotype and reports both distinct-rule and Shannon-effective diversity at* $T = 70$*; full-population seed* 0*,* $L = 256$*. The fields sustain* 78–121 *rules and carry coherent diagonal structure at a range of scales. These examples motivated the candidate diversity coordinate, but the conservative-flow control in Figure 16 shows that diversity itself does not account for their propagation. Continued in (b).*

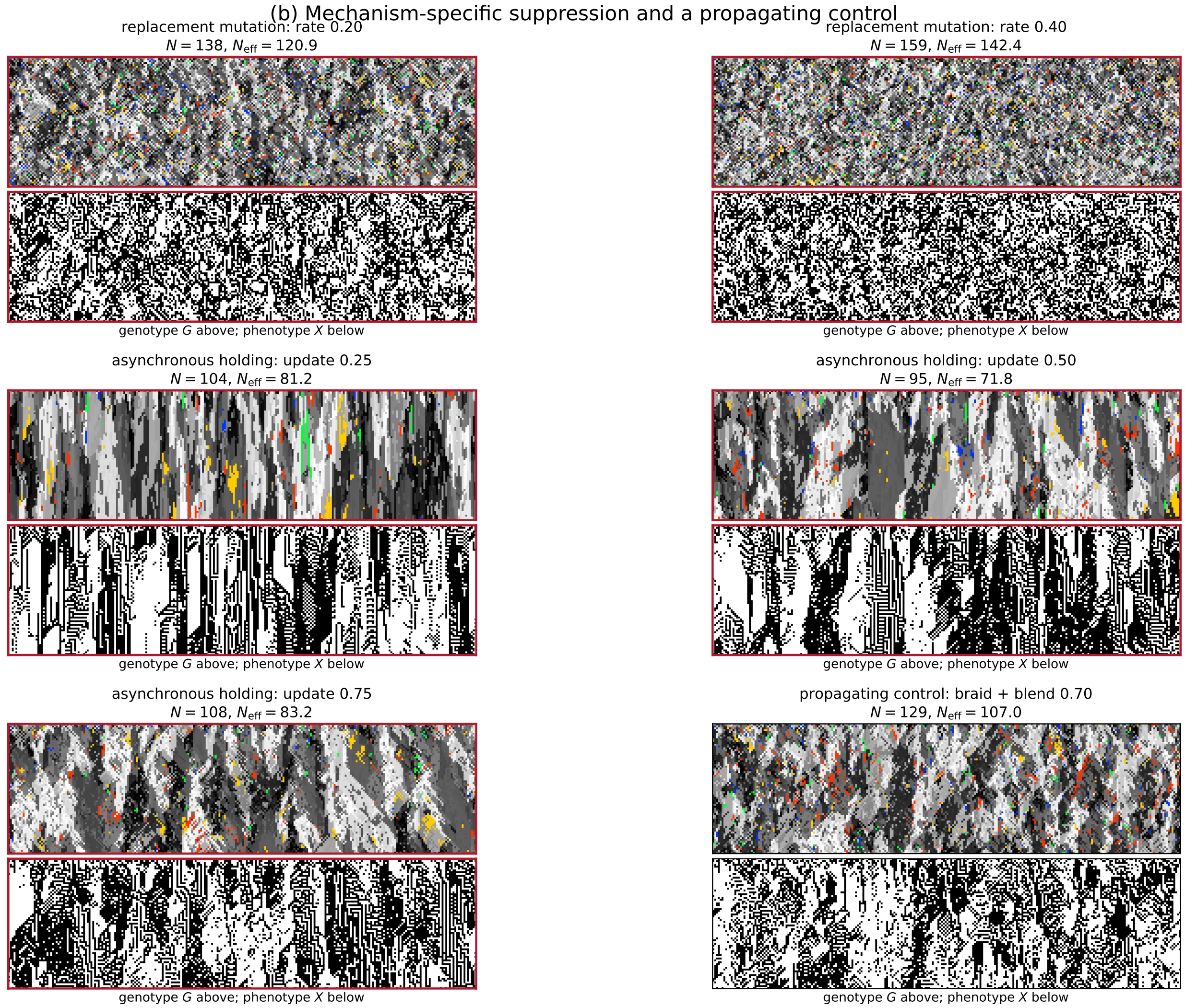

Figure S3.2

(b) Two mechanism-specific suppressions. *Continued from (a); suppressed configurations are outlined in red. Replacement mutation at rates* 0.20 *and* 0.40 *turns the genotype into progressively finer, unstructured speckle. Asynchronous holding at update fractions* 0.25, 0.50, *and* 0.75 *instead produces vertical persistence whose strength changes continuously with the dial. The unoutlined braid-plus-blend control at bottom right propagates (*3.88 *cells) and recovers the diagonal texture of (a). The expanded population makes the distinction visible as a family resemblance rather than a single selected contrast: these are genuine failure modes of their respective mechanisms, not evidence for suppression by high diversity.*